SoRoSim

A MATLAB Toolbox for Hybrid Rigid-Soft Robots
Based on the Geometric Variable-Strain Approach

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The Challenge in Soft Robotics

Current Limitations

- Limited modeling tools for soft robots
- FEM approaches are computationally heavy
- 1000s of degrees of freedom required
- Slow computation times
- Difficulty modeling hybrid rigid-soft systems
- Lack of unified framework
- Time-consuming setup for each robot

Research Needs

- Medical robotics
- Continuum robots & catheters
- Surgical instruments
- Soft actuators (LCE, pneumatic)
- Haptic devices and interfaces
- Underwater/bio-inspired robots
- Real-time control implementation

What is SoRoSim?

A user-friendly MATLAB toolbox that integrates the Geometric Variable-Strain (GVS) model of Cosserat rods for efficient soft robot simulation

Unified Framework
Models soft, rigid, and hybrid robotic systems

Efficient Computation
Faster than real-time using minimal DoF

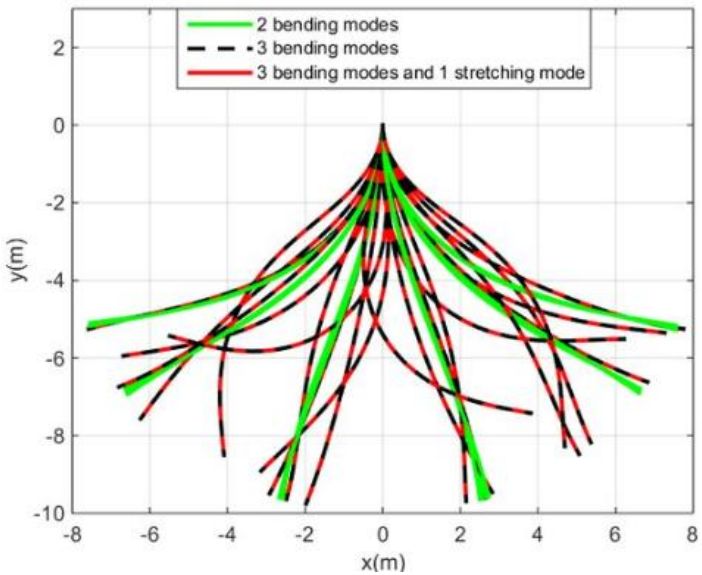
Open-Source
Free on GitHub with full documentation

User-Friendly
GUI-based design with MATLAB integration

How SoRoSim Works - The GVS Approach

Cosserat Rod Theory

- Models soft bodies as 1D continuum mechanics objects
- Accounts for all 6 deformation modes:
  - 3 rotational: torsion, bending (y/z axes)
  - 3 translational: extension, shear (y/z axes)



Geometric Variable-Strain (GVS) Method

Key Innovation:

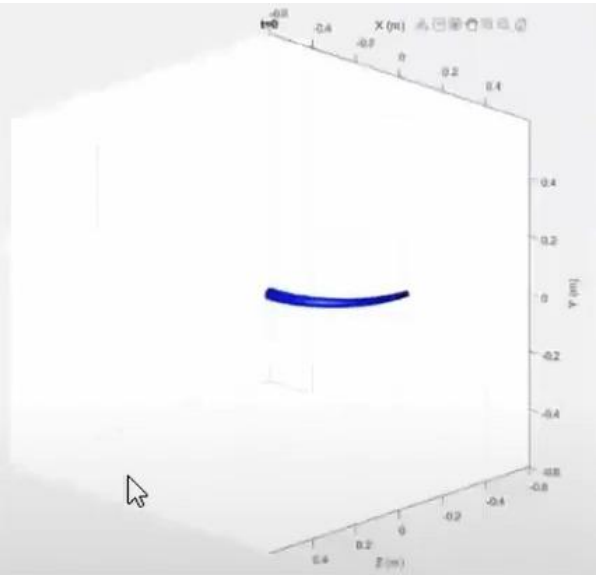
- Parameterizes strain directly
- Instead of storing the exact position & orientation of every node, you store how much it bends/twists/stretches along its length.
- Far fewer variables → lighter, faster, great for control & simulation
- Geometrically exact formulation maintains accuracy

Result: Same accuracy as FEM with 100× fewer variables → 3000× faster computation

Key Capabilities & Features

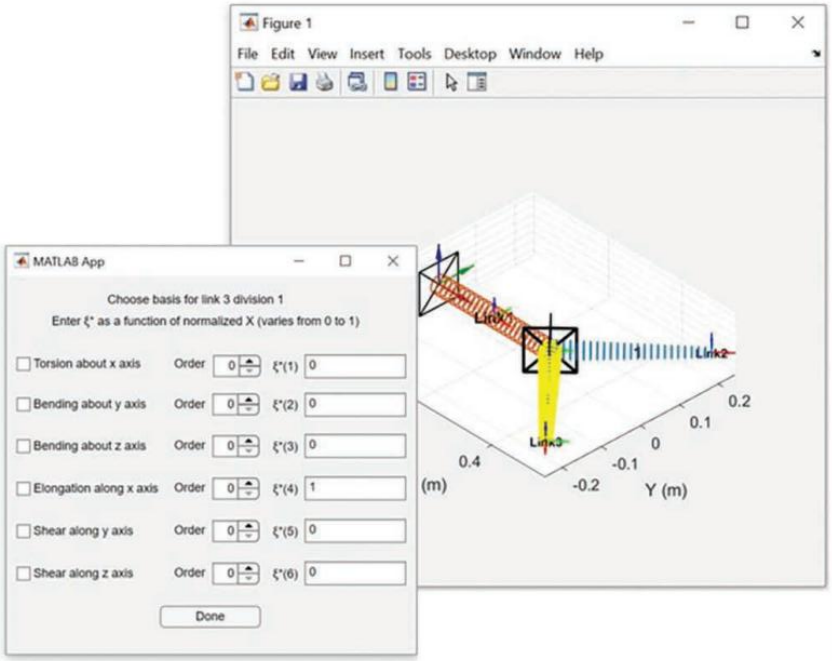
Modeling Features

- 9 joint types (revolute, prismatic, spherical, universal, etc.)
- Open, branched, and closed-chain configurations
- Custom material models (linear/nonlinear)
- Variable cross-sections along length
- 6 deformation modes per soft link
- Multi-division soft links



Analysis Types

- Static equilibrium analysis
- Dynamic simulation with contact
- Design optimization
- Inverse dynamics control
- Energy analysis
- Custom forces and actuation



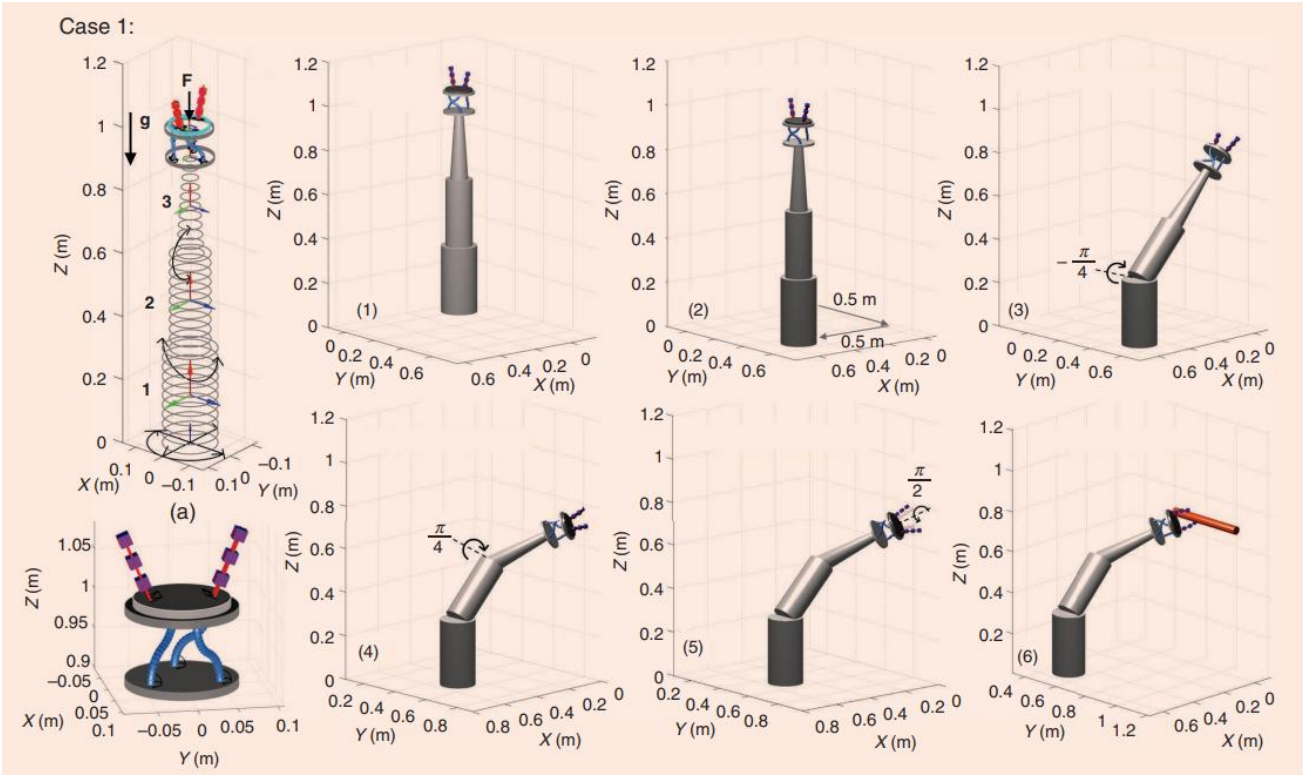
Static Equilibrium of Hybrid Robotic Arm

System Specifications

- 11 links (rigid and soft)
- 39 degrees of freedom
- 7 actuation inputs
- Multiple joint types

Performance

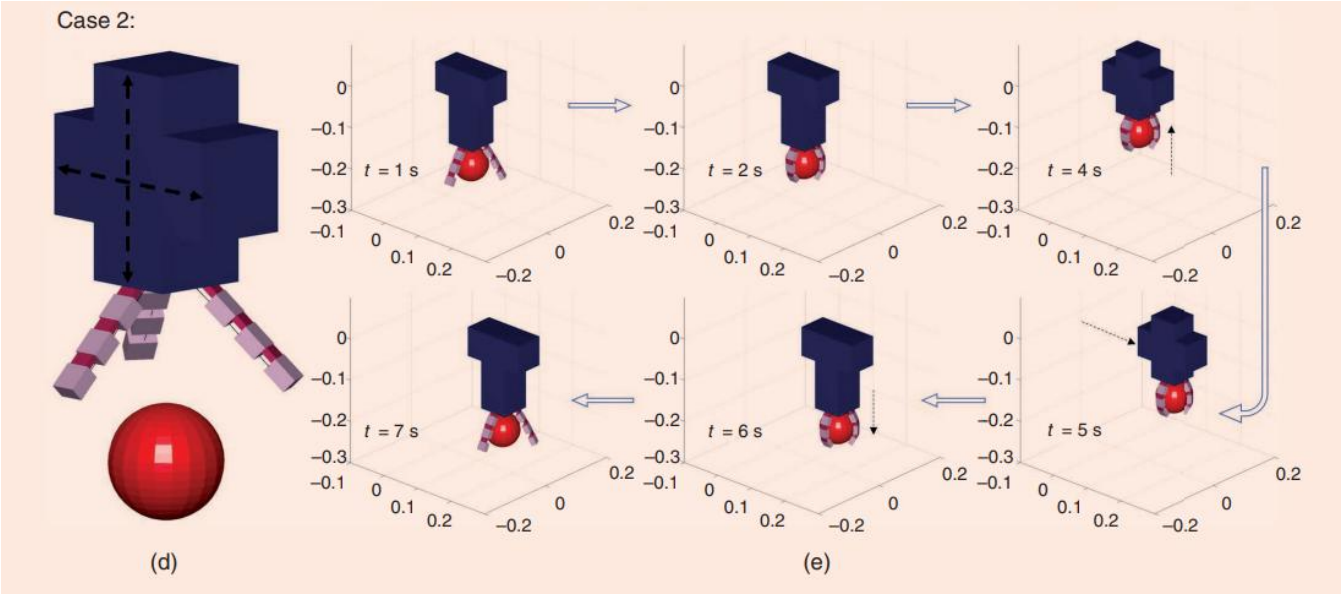
- Static equilibrium: 1.2 seconds

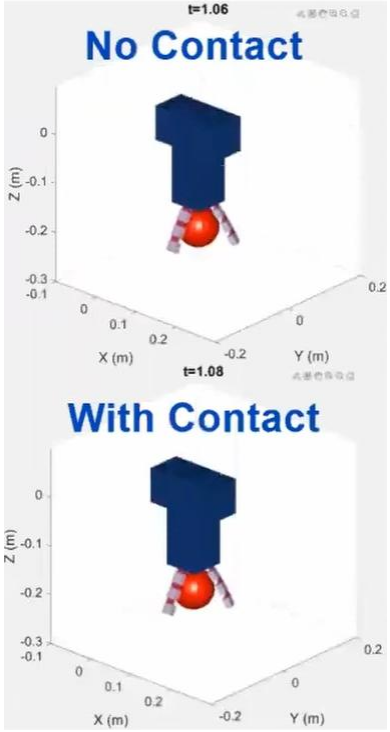


Contact Dynamics of Soft Gripper

Modeling Custom Forces

- SoRoSim has a CustomExtForce.m MATLAB Function
- Quantities such as Jacobian J, screw velocity h, and transformation matrices g are passed as inputs into this function.
- The user can apply a custom external force as a function of these quantities.

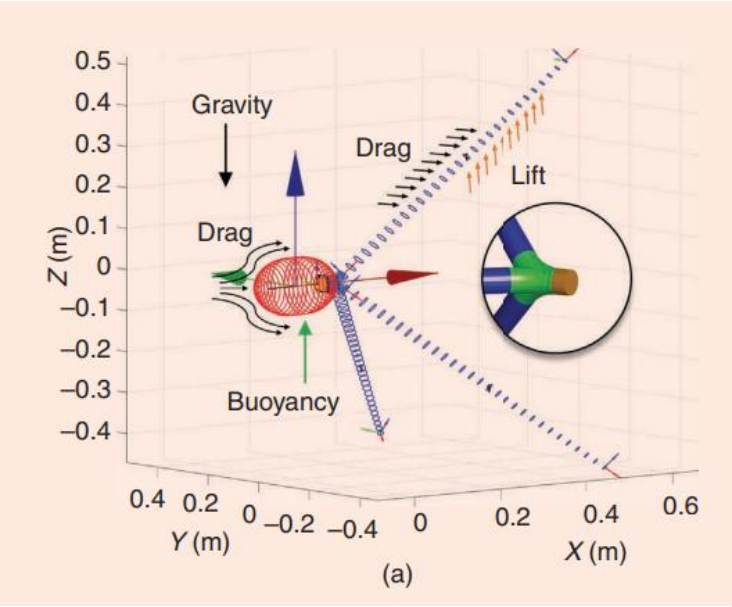


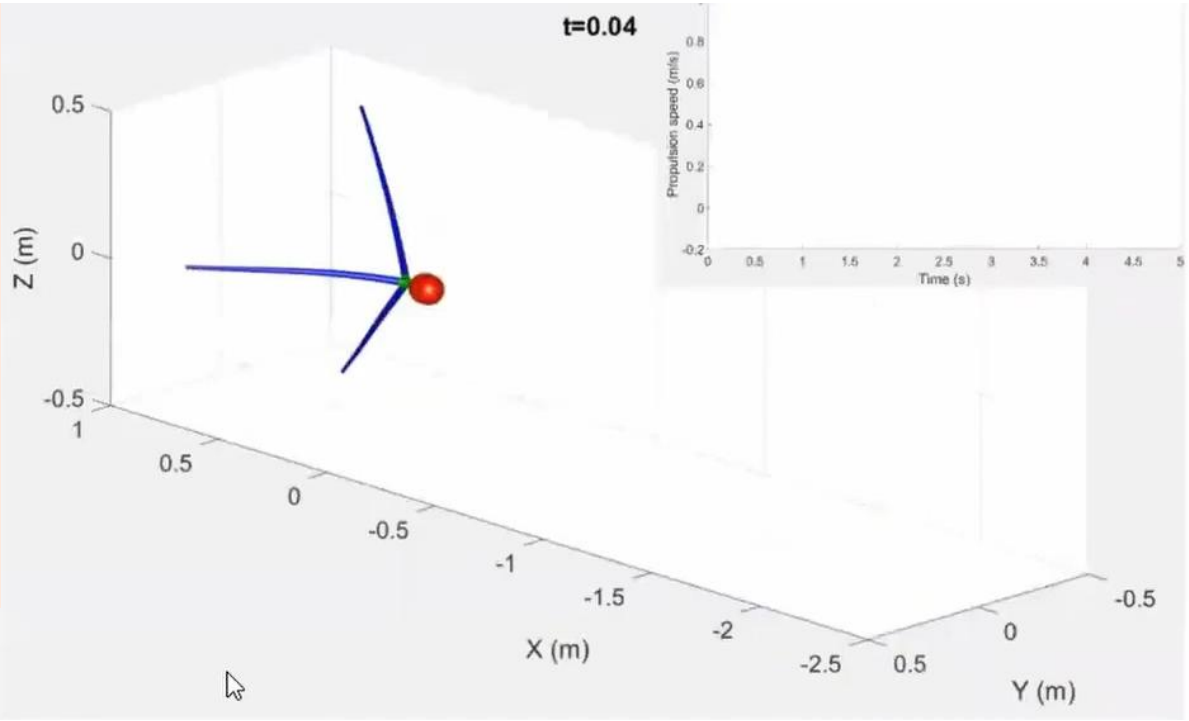


Underwater Robots

System Specifications

- SoRoSIm can simulate the fluid–robot interaction forces.
- We can also define custom fluid forces using CustomExtForce.m function.
- Thrust is generated by the soft propeller by providing an actuation input of θ = 2πt



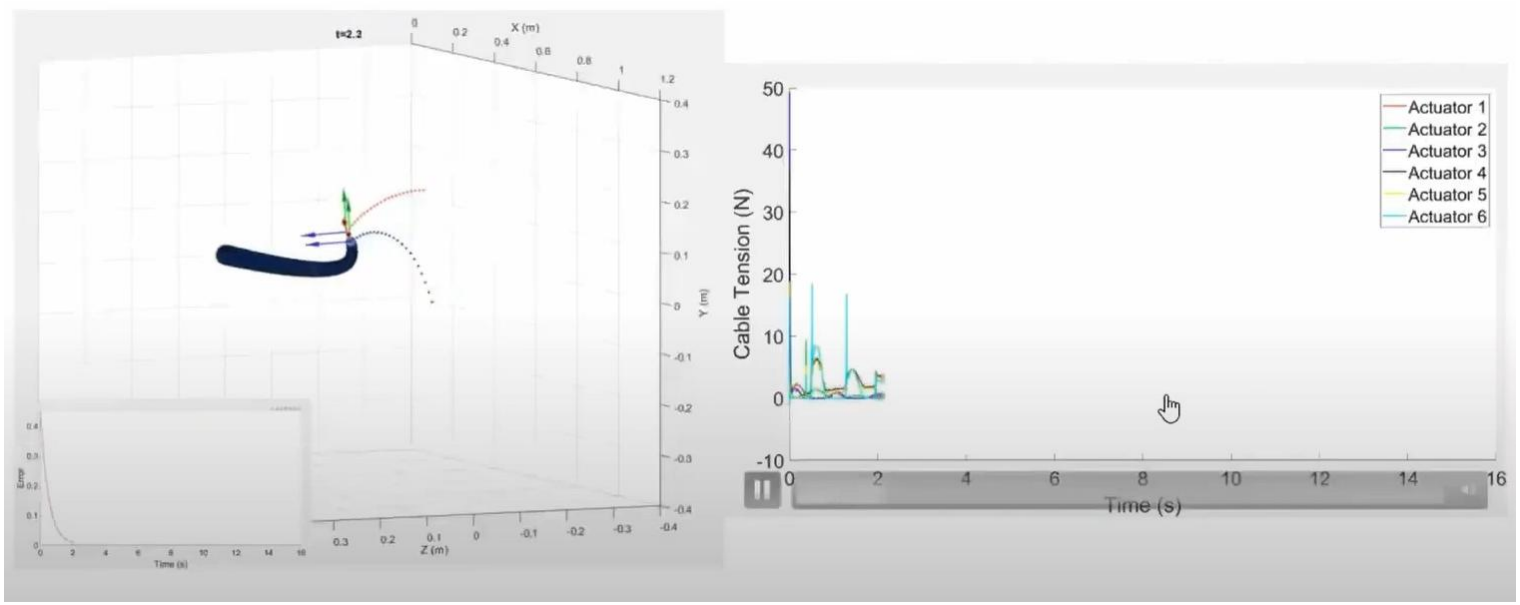


Tip Pose Trajectory Tracking

System Specifications

- This soft manipulator is actuated using six linearly independent cables.
- From the dynamics equation of the manipulator tip, solve for the actuator strength, u.
- Apply a proportional derivative (PD) controller (8).
- Toolbox then incorporates the control law.
- We use the “lsqlin” function of MATLAB to compute the actuator strength given by (8).
- Trajectory tracking: <4s convergence.

$$\boldsymbol{u} = (\boldsymbol{J}_t\boldsymbol{M}^{-1}\boldsymbol{B})^{\dagger}[\dot{\bar{\boldsymbol{\eta}}}_t + K_d(\bar{\boldsymbol{\eta}}_t - \boldsymbol{\eta}_t) + K_p(\log(\boldsymbol{g}_t^{-1}\bar{\boldsymbol{g}}_t))^{\vee} + \boldsymbol{J}_t\boldsymbol{M}^{-1}((\boldsymbol{C}+\boldsymbol{D})\dot{\boldsymbol{q}} + \boldsymbol{K}\boldsymbol{q} + \boldsymbol{Q}) - \dot{\boldsymbol{J}}_t\dot{\boldsymbol{q}}] \qquad (8)$$



Design Optimization

System Specifications

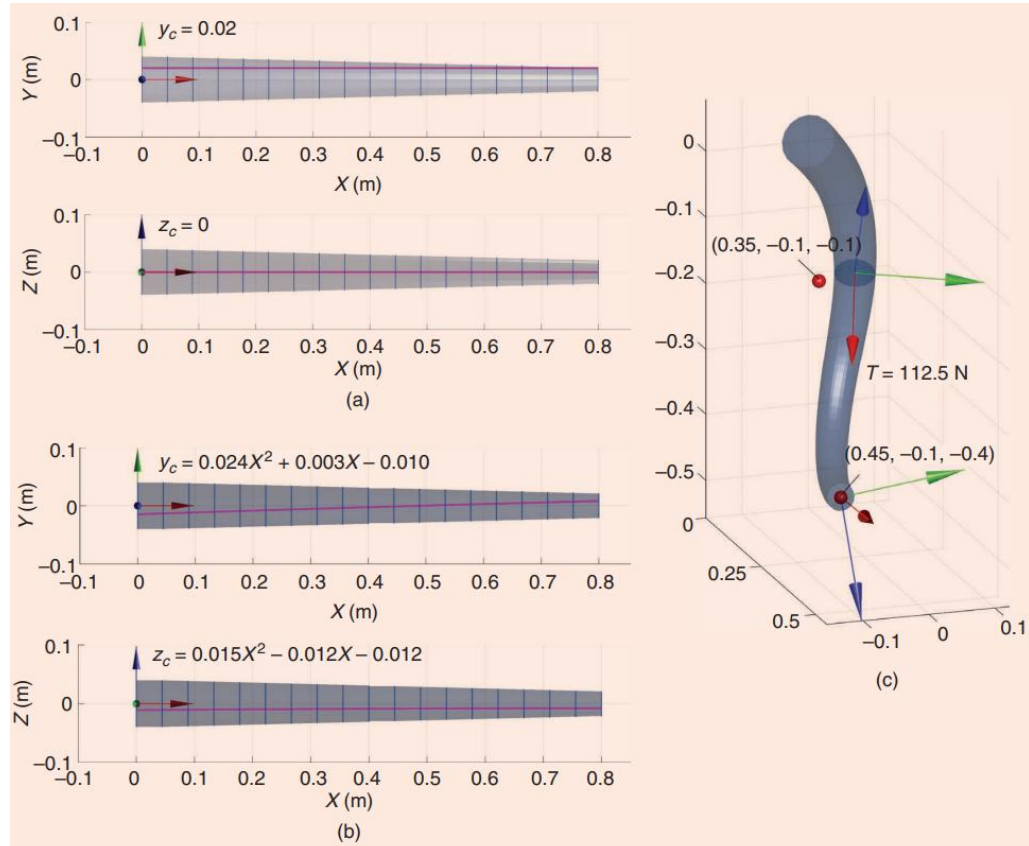
- We set up an optimization problem to find an appropriate cable path and cable tension.
- We are minimizing the error in the manipulator end, and middle point coordinates with desired positions at static equilibrium.
- The objective function is given:

$$\Omega = \| \bar{\boldsymbol{r}}_{\text{mid}} - \boldsymbol{r}_{\text{mid}} \| + \| \bar{\boldsymbol{r}}_{\text{end}} - \boldsymbol{r}_{\text{end}} \| \qquad (7)$$

- We use the MATLAB "patternsearch" algorithm to find the optimal values of variables at the local minimum of the objective function
- The optimal solution is given by:

$$y_c = 0.024X^2 + 0.003X - 0.010, \; z_c = 0.015X^2 - 0.012X$$

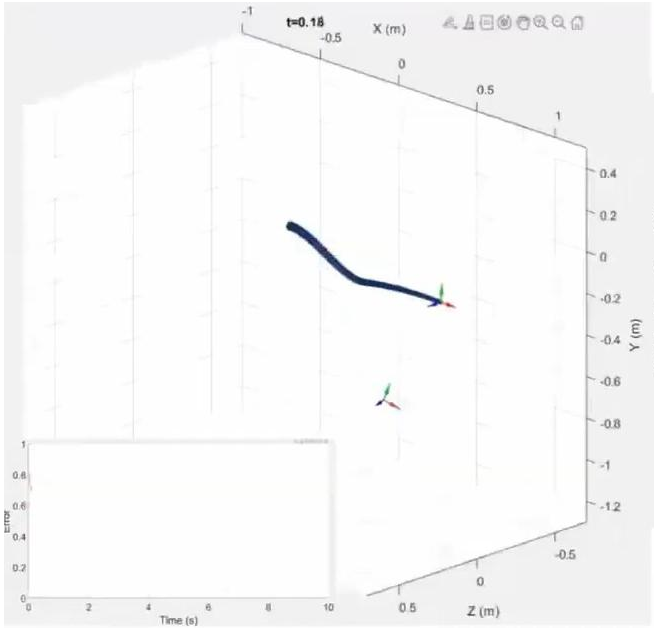
and $T = 112.5$ N.

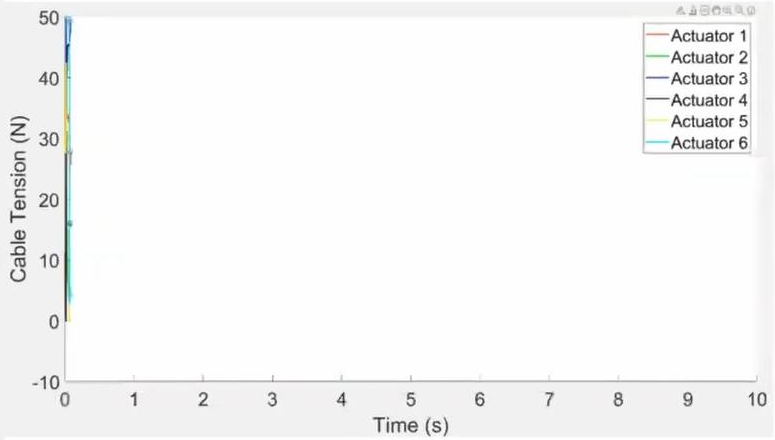


Tip Pose Regulation Under Gravity

System Specifications

- Attempt to control the position and orientation of the manipulator tip based on a fixed reference frame under the influence of gravity.
- Using the PD controller given by (8), we can overcome the dynamic response due to gravity.
- The error is quickly reduced to zero in fewer than 5 s.





Performance & Validation

Toolbox performs in real time or faster for most of the examples

DOF Reduction: 100:1 vs FEM

The newest version, Differential_SoRoSim, uses analytical derivatives of the governing equations of GVS for even faster analysis over 2.5× faster than real time!

Validation Studies

✓ Matches analytical solutions (cantilever beam tests)
✓ Agrees with ANSYS FEM (2% error, 100× fewer DOF)
✓ Energy conservation verified

Table 2. A summary of the toolbox performance.

Example	N	N_{dof}	$\xi(O)$	$Z(O)$	N_{eva}	N_s (Number)/ t_{max} (s)	Solver	t_s (s)
"Test 1: Fixed–Free Beam With a Follower Tip Force" section	1	5	4	4	16	130	*fsolve*	5.5
"Test 2: FEM Study of a Fixed–Fixed L-Shaped Beam" section	2	30	2	4	20	1	*fsolve*	0.5
"Test 3: Dynamics of a Flexible Flying Rod" section	1	15	2	4	13	7	*ode45*	1
"Test 4: Energy Balance in SoRoSim Dynamics" section	1	12	3	4	15	5	*ode15s*	1.5
"Test 4: Energy Balance in SoRoSim Dynamics" section*	1	12	3	2	15	5	*ode113*	6
"Case 1: Static Equilibrium" section	11	39	0	4	143	5	*fsolve*	6
"Case 2: Contact Dynamics" section	6	21	0	4	96	9	*ode15s*	27
"Underwater Robotics" section	6	20	1	2	30	5	*ode113*	5
"Design Optimization" section	1	12	1	4	15	2,887	*fsolve, patternsearch*	93
"Case 1: Tip Pose Trajectory Tracking" section	1	6	1	4	15	16	*ode1*†	5
"Case 2: Tip Pose Regulation Under Gravity" section	1	6	1	4	15	10	*ode1*†	4

Static simulations are highlighted in gray.
*With undamped ($\boldsymbol{D} = \boldsymbol{0}$) dynamics.
†With a time step of 0.01 s.

Conclusions & Getting Started

Why SoRoSim Accelerates Research

✅ Rapid Prototyping: Concept to simulation in minutes
✅ Complex Designs: Model intricate medical devices
✅ Optimization Ready: Find problem-specific solutions
✅ Control Testing: Implement algorithms efficiently
✅ Publication Ready: Validated and cited framework

📦 GitHub: https://github.com/SoRoSim/SoRoSim